

H524 Introduction to Biostatistics

Week 6: Power, Two-Sample Tests, and ANOVA

Homework Assignment

Fall 2025

Due Date: End of Week 6

Total Points: 10

Format: Submit through Canvas

Time Limit: 60 minutes

Instructions

- Show all work for full credit
- Round numerical answers to 2 decimal places unless otherwise specified
- Include R code where specified
- You may use R to perform calculations

1 Numerical Questions (10 points total)

Question 1: Sample Size for Power (5 points)

A researcher is planning a study to test whether a new medication reduces systolic blood pressure. Based on pilot data, the population standard deviation is estimated to be $\sigma = 20$ mmHg. The researcher wants to detect a mean reduction of 10 mmHg from the population mean of 145 mmHg.

Using a significance level of $\alpha = 0.05$ (two-sided) and desired power of 0.80, calculate the required sample size using R's `power.t.test()` function for a one-sample t-test.

Show your R code:

What is the required sample size (round UP to the nearest whole number)?

Answer: _____ subjects

Question 2: Two-Sample t-Test and ANOVA (5 points)

A researcher compares the effectiveness of three different exercise programs on weight loss (in kg) over 8 weeks. The data are:

Program A	Program B	Program C
4.2	6.1	7.8
3.8	5.8	8.2
4.5	6.4	7.5
4.1	5.9	8.0
4.3	6.2	7.9

Part A: Perform a one-way ANOVA to test if there is a significant difference in mean weight loss across the three programs. Use $\alpha = 0.05$.

Show your R code to create the data and perform ANOVA:

What is the F-statistic (round to 2 decimal places)?

Answer: F = _____

Part B: Compare ONLY Program A vs. Program C using a two-sample t-test (assume equal variances, two-sided test).

Show your R code:

What is the p-value (round to 4 decimal places)?

Answer: p-value = _____